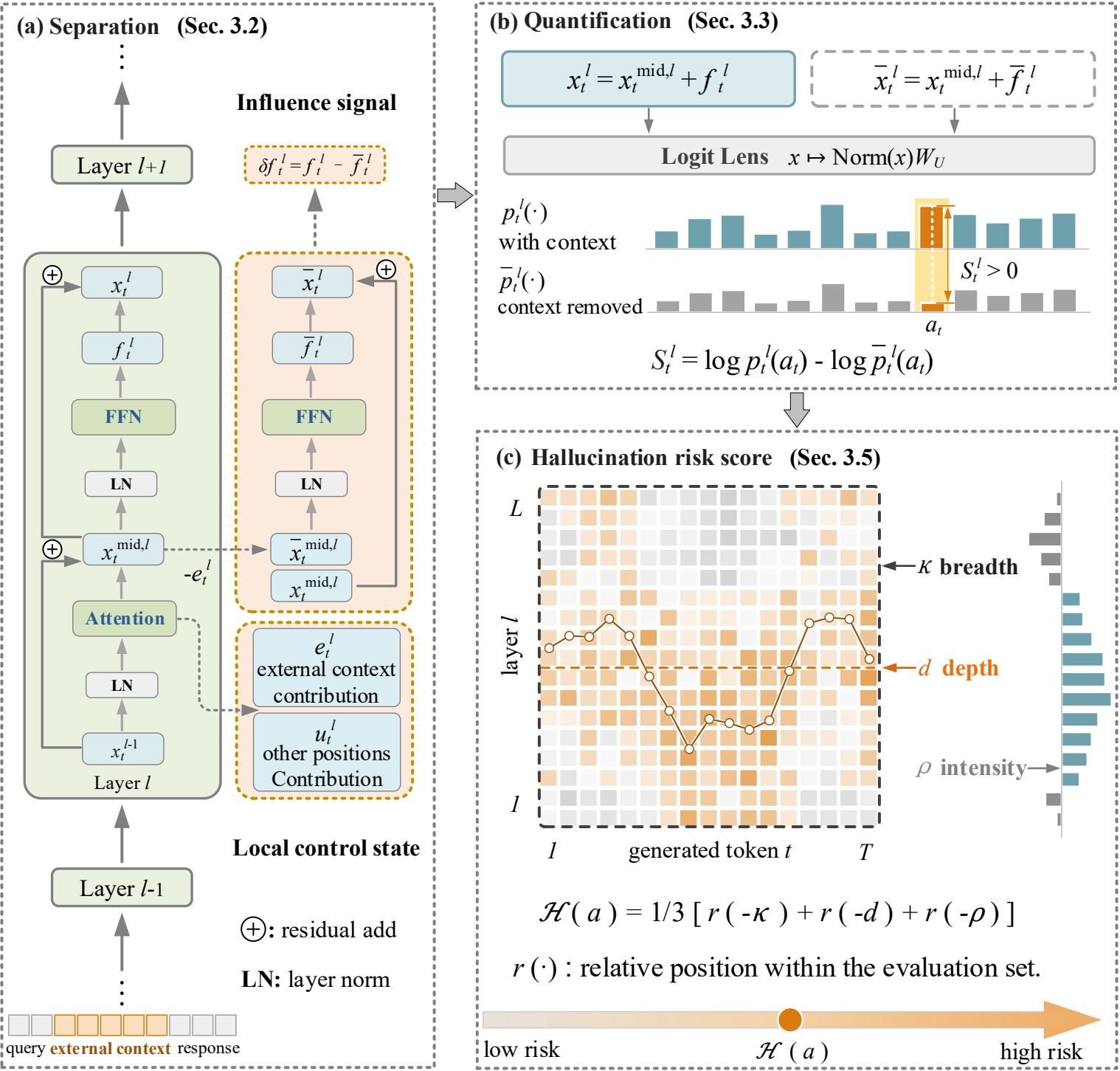

(a) Separation (Sec. 3.2)
Influence signal
Layer l+1
$\delta f_t^l = f_t^l - \bar{f}_t^l$
x_t^l
f_t^l
FFN
LN
$x_t^{\text{mid},l}$
$-e_t^l$
Attention
LN
x_t^{l-1}
Layer l
$\bar{x}_t^l$
$\bar{f}_t^l$
FFN
LN
$\bar{x}_t^{\text{mid},l}$
$x_t^{\text{mid},l}$
e_t^l external context contribution
u_t^l other positions Contribution
Local control state
Layer l-1
⊕: residual add
LN: layer norm
query external context response
(b) Quantification (Sec. 3.3)
$x_t^l = x_t^{\text{mid},l} + f_t^l$
$\bar{x}_t^l = x_t^{\text{mid},l} + \bar{f}_t^l$
Logit Lens $x \mapsto \text{Norm}(x)W_U$
$p_t^l(\cdot)$ with context
$\bar{p}_t^l(\cdot)$ context removed
$S_t^l > 0$
a_t
$S_t^l = \log p_t^l(a_t) - \log \bar{p}_t^l(a_t)$
(c) Hallucination risk score (Sec. 3.5)
L
layer l
1
1
generated token t
T
κ breadth
d depth
ρ intensity
$\mathcal{H}(a) = 1/3\,[\,r(-\kappa) + r(-d) + r(-\rho)\,]$
$r(\cdot)$: relative position within the evaluation set.
low risk
$\mathcal{H}(a)$
high risk